

E2BCA1T3 – E – 25 – 2087



SECOND SEMESTER B.C.A. (REVISED NEP) DEGREE
EXAMINATION, JULY 2025
PROBABILITY AND STATISTICS (DSC – 3)
Theory

Time : 3 Hours]

[Max. Marks : 80

- Instructions :** i) Answer Part – A, Part – B, Part – C questions.
ii) **Calculators** are allowed.

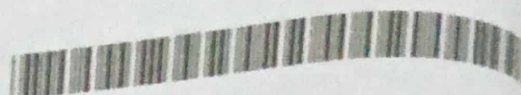
PART – A

Answer **all** questions. **Each** carries **2** marks.

(10×2=20)

1. Define probability with an example.
2. What is sample space and events in probability ?
3. Find the probability of getting an Ace when a card is drawn from a pack of playing cards.
4. Define Estimation.
5. What is population ? Give an example.
6. Define variance and standard deviation.
7. Define Hypothesis.
8. What is critical region ?
9. List any two types of correlation.
10. Define Regression.

[P.T.O.]



PART – B

(6×5=30)

Answer **any 6** questions. **Each** carries **5** marks.

11. A box has 100 tickets marked with numbers from 1 to 100. One ticket is drawn at random from the box. Find the probability that the number on drawn ticket is

i) less than 20

ii) Multiple of 5.

12. Define Random variables in probability distribution, explain its types.

13. Explain the properties of a good estimator.

14. A random sample of 100 observations yields sample mean $\bar{X} = 150$ and sample variance $s^2 = 400$. Compute 95% confidence interval for population mean. (Assume value of $Z = 1.96$)

15. Explain Null Hypothesis and Alternate Hypothesis with an example.

16. Write properties of Chi-Square distribution.

17. Calculate rank co-efficient of correlation for the following data.

X	1	2	3	4	5	6	3	10	15	20
Y	15	25	10	20	8	16	17	30	30	31

18. Compute co-efficient of correlation for the following data.

X	1	2	3	4	5
Y	8	9	11	12	10

PART – C

Answer **any 3** questions. **Each** carries **10** marks.

(3×10=30)

19. State and prove Baye's theorem.

20. a) Explain Student's T-distribution.

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b) Define Standard Error and Acceptance Region.

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21. a) Write steps in carrying out Hypothesis testing.

b) Following information is available with respect to 2 brands of bulbs.

	Brand A	Brand B
Mean life (hrs)	1300	1248
SD	82	93
Sample size	100	100

Which brand should be preferred at 5% level of significance ? (5+5=10)

22. From the following data find regression lines.

X	5	4	6	8	12
Y	1	5	10	6	8

Estimate the value of X when $Y = 15$.
